

- At a given exercise intensity in the heat, cardiac output may remain reasonably constant or decrease slightly at higher intensities, as a gradual upward drift in heart rate helps offset a lower stroke volume.
 - Prolonged heavy sweating can lead to dehydration and excessive electrolyte loss. To compensate, increased release of aldosterone and ADH enhances sodium and water retention.
 - Sweating rates as high as 4 L per hour have been observed in highly trained, well-acclimated athletes, but these rates cannot be sustained for more than several hours. Maximal daily sweating rates can be in the range of 10 to 15 L but only with adequate fluid replacement.
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Health Risks During Exercise in the Heat

Despite the body's defenses against overheating, excessive heat production by active muscles, heat gained from the environment, and conditions that prevent the dissipation of excess body heat may elevate the internal body temperature to levels that impair normal cellular functions. Under such conditions, excessive heat gains pose a risk to one's health, as we highlighted in the opening anecdote of this chapter. Air temperature alone is not an accurate index of the total physiological stress imposed on the body in a hot environment. Six variables must be taken into account:

- Metabolic heat production
- Air temperature
- Ambient water vapor pressure (humidity)
- Air velocity
- Radiant heat sources
- Clothing

All of these factors influence the degree of heat stress that a person experiences. The contribution of each factor to the total heat stress under various environmental conditions can be predicted mathematically using advanced heat balance equations.

RESEARCH PERSPECTIVE 12.2

Tattoos and Sweating

The practice of tattooing dates back to ancient civilization but has become increasingly popular in recent times. Approximately 45 million Americans have at least one tattoo, and 40% of young adults (20-40 years) have one or more tattoos compared to only 10% of older adults.¹⁰ Tattoos are particularly popular among college and professional athletes as well as military personnel, who are regularly physically active and rely on increasing skin blood flow and eccrine sweating to dissipate metabolic heat produced during physical activity.

The tattooing process involves puncturing the skin with a cluster of needles to deposit dye 3 to 5 mm below the skin's surface into the dermal layer. This process initiates an inflammatory response within the dermal layer of the skin, which contributes to the permanent color incorporated in the skin. The dermal layer of the skin is composed of collagen fibers, nerves, blood vessels, and eccrine sweat glands. Eccrine sweat glands draw fluid from the extracellular fluid into the secretory coils. As isosmotic primary sweat is extruded through the skin surface, sodium chloride is reabsorbed, resulting in a very low sweat sodium concentration. Because the eccrine sweat glands are located in the dermal layer, where the primary effects of tattooing occur, it is possible that tattooing will affect the sweat gland function (sweat production or sodium reabsorption) in the tattooed skin.

A recent study conducted at Alma College in Michigan examined the sweating response in tattooed compared to nontattooed skin.⁷ In this study, the researchers stimulated sweating by delivering pilocarpine nitrate (a pharmacological agonist that stimulates muscarinic receptors on the sweat gland) to a 5.2 cm area of tattooed skin and a 5.2 cm area of nontattooed skin on the contralateral anatomical location on the same subjects. The researchers chose to measure the contralateral anatomical location to account for regional differences in sweat rate across the body. They measured the sweat rate by weighing an absorbent disk placed on the skin before and after the sweating stimulus, and collected the sweat from each site to measure sodium concentration by flame photometry later. The study team found that sweat rate in the tattooed skin was significantly lower compared to the nontattooed skin and that the sodium concentration was much higher in the tattooed skin. These data suggest that tattooing alters the function of the sweat gland in the area of skin that is tattooed. This study is the first of its kind and will likely open the door for future research exploring the mechanisms by which these alterations occur and whether there is any way to restore sweat gland function in tattooed skin.

An individual exercising on a bright, sunny day with an air temperature of 23 °C (73.4 °F) and no measurable wind experiences considerably more heat stress than someone exercising in the same air temperature but under cloud cover and with a slight breeze. At

temperatures above skin temperature, which is normally 32 to 33 °C (92 °F), radiation, conduction, and convection substantially add to the body's heat load rather than acting as avenues for heat loss. How, then, can we estimate the amount of heat stress to which an individual may be exposed?

Measuring Heat Stress

It has become common to hear about the heat index on local weather channels. The heat index, a complex equation involving air temperature and relative humidity, is a measure of how hot it *feels*, that is, how we perceive the heat. However, the heat index does not do a good job of reflecting the physiological stress on humans, so its use is limited in exercise physiology. Through the years, efforts have been made to quantify atmospheric variables into a single index that would reflect the physiological heat stress on an individual. In the 1970s, **wet-bulb globe temperature (WBGT)** was devised to simultaneously account for conduction, convection, evaporation, and radiation (see [figure 12.8](#)). It is based on three different thermometer readings and provides a single temperature reading to estimate the cooling capacity of the surrounding environment.

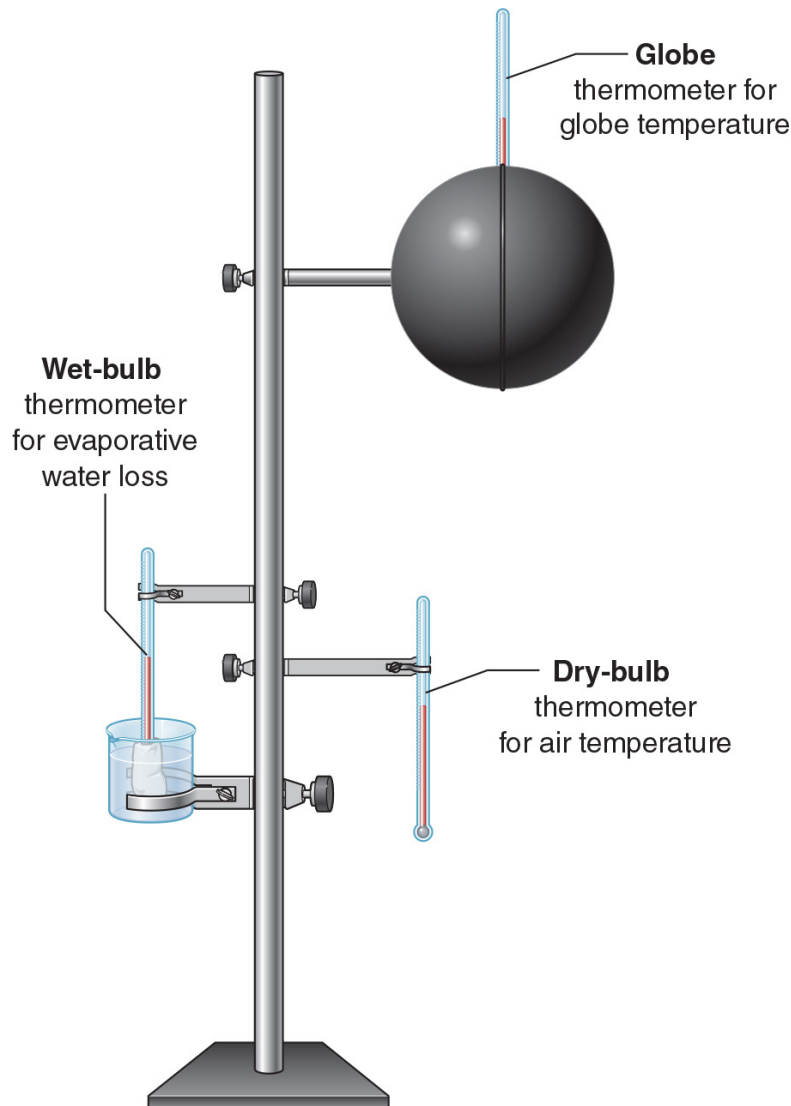


FIGURE 12.8 A wet-bulb globe temperature stand showing the three separate thermometers for air (dry-bulb) temperature, wet-bulb temperature reflecting the cooling effect of evaporation, and globe temperature, which measures additional effects of radiant heat.

The dry-bulb temperature (T_{db}) is the actual air temperature one would measure with a typical thermometer. A second thermometer has a bulb that is kept moist by a wetted cotton “sock” dipped in distilled water. As water evaporates from this wet bulb, its temperature (T_{wb}) will be lower than the T_{db} , reflecting the effect of sweat evaporating from the skin. The difference between the wet- and dry-bulb temperatures indicates the environment’s capacity for cooling by evaporation. In still air with 100% relative humidity, these two bulb temperatures are the same because evaporation is impossible. Lower ambient water vapor pressure and increased air

movement promote evaporation, increasing the difference between these two bulb temperatures. The third thermometer, placed inside a black globe, typically shows a temperature higher than T_{db} as the globe, painted a flat black, maximally absorbs radiant heat. Thus, its temperature (T_g) is a good indicator of the environment's radiant heat load.

The temperatures from these three thermometers can be combined into the following equation to estimate the overall atmospheric challenge to body temperature in outdoor environments:

$$WBGT = 0.1 T_{db} + 0.7 T_{wb} + 0.2 T_g$$

The fact that the coefficient for T_{wb} is the largest reflects the importance of sweat evaporation in the physiology of heat exchange. Also note that WBGT reflects only the environment's impact on heat stress and is most effectively used along with a measure or estimate of metabolic heat production. Clothing further influences heat stress.

Wet-bulb globe temperature, as an index of **thermal stress**, is now routinely used by coaches, team physicians, and athletic trainers to anticipate the health risks associated with athletic practices and competitions in thermally stressful environments.

Heat-Related Disorders

Exposure to the combination of external heat stress and metabolically generated heat can lead to three heat-related disorders (see [figure 12.9](#)): heat cramps, heat exhaustion, and heatstroke.

Heat Cramps

Heat cramps, the least serious of the three heat disorders, are characterized by severe and painful cramping of large skeletal muscles. They involve primarily the muscles that are most heavily used during exercise, and such instances of athletes "locking up" are different from cramps everyone has experienced in small muscles. Heat cramps are brought on by sodium losses and dehydration that accompany high rates of sweating, and thus are most common in heavy sweaters who lose a lot of sodium in their sweat. (A common misconception is that potassium is involved in cramping and that eating potassium-rich foods like bananas will prevent heat cramps.) Heat cramps can be prevented or minimized in susceptible athletes

by proper hydration practices involving liberal salt intake with foods and in beverages consumed during exercise. Treatment for these cramps involves moving the stricken individual to a cooler location and administering a saline solution, either orally or intravenously if necessary.

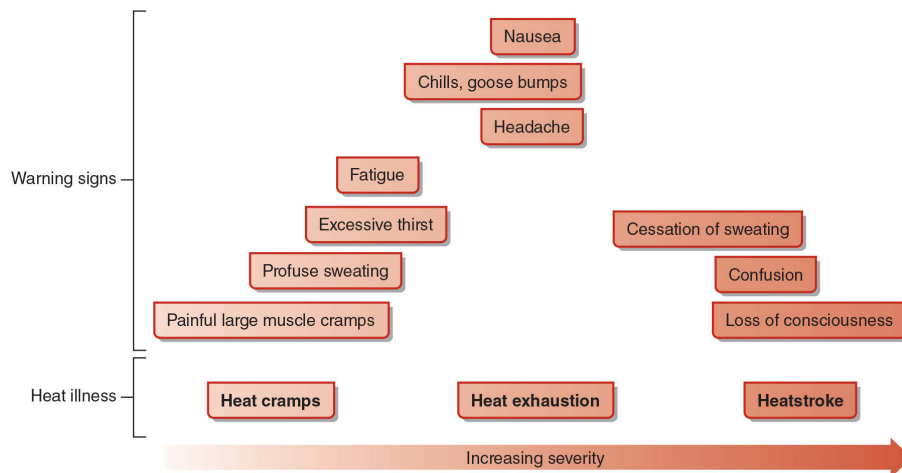


FIGURE 12.9 Schematic diagram of heat illnesses of increasing severity, showing some of the typical warning signs that accompany these illnesses. Not all of these warning signs are present in every case of heat illness, nor do they always occur in this, or any predictable, order. Any of the three heat illnesses shown here can occur suddenly without prior symptoms; that is, there is not always a progression from heat exhaustion to heatstroke.

Adapted by permission of All Sport, Inc.

Heat Exhaustion

Heat exhaustion typically is accompanied by such symptoms as extreme fatigue, dizziness, nausea, vomiting, fainting, and a weak, rapid pulse. It is caused by the cardiovascular system's inability to adequately meet the needs of the body as it becomes severely dehydrated. Recall that during exercise in heat, active muscles and skin compete for a share of a limited, and decreasing, blood volume. Heat exhaustion may result when these simultaneous demands cannot be met, and it typically occurs when blood volume decreases as a result of excessive fluid loss from profuse sweating. A second form of heat exhaustion, from sodium depletion, is rare in athletes. Therefore, heat exhaustion can be thought of as a syndrome of dehydration and should be treated as such.

With heat exhaustion, the thermoregulatory mechanisms are functioning but cannot dissipate heat quickly enough because

insufficient blood volume is available to allow adequate blood flow to the skin. Although the condition often occurs during moderate to heavy exercise in the heat, it is not necessarily accompanied by extremely high core temperatures. Some people who collapse from heat exhaustion have core temperatures well below 39 °C (102.2 °F). People who are unfit or not acclimated to the heat are more susceptible than others to heat exhaustion.

Treatment for victims of heat exhaustion involves rest in a cooler environment with their feet elevated to facilitate return of blood to the heart. If the person is conscious, administration of salt water is usually recommended. If the person is unconscious, medically supervised intravenous administration of saline solution is recommended.

Heatstroke

As we saw in our opening story, **heatstroke** is a life-threatening heat disorder that requires immediate medical attention. Heatstroke is caused by failure of the body's thermoregulatory mechanisms. It is characterized by

- an increase in internal body temperature to a value exceeding 40 °C (104 °F) and
- confusion, disorientation, or unconsciousness.

The latter element—altered mental status—is the key to recognizing impending heatstroke because neural tissues in the brain are particularly sensitive to extreme heat. In heatstroke, cessation of active sweating may also occur, but sweat may remain on the skin. The notion that heatstroke is always accompanied by dry, red skin is outdated, and this sign should never be used to distinguish heatstroke from heat exhaustion.

If heatstroke is left untreated, core temperature will continue to rise, progressing to coma and ultimately death. Appropriate treatment always involves cooling the body as rapidly as possible. In the field, this can best be accomplished by immersing the victim—as much of the body as possible excluding the head—in a bath of cold water or ice water. While cold-water immersion provides the fastest cooling rates, in cases in which cold or ice water is not available,

temperate-water immersion is the next best option. Where immersion is not feasible, the combination of wrapping the entire body in cold, wet sheets and fanning vigorously may be used. Cooling methods that place ice bags on small areas, like the armpits, neck, and groin, are not effective in rapidly lowering core temperature because of the small surface area covered.

For the athlete, heatstroke is not just a problem associated with extreme conditions. Studies have reported rectal temperatures above 40.5 °C (104.9 °F) in marathon runners who successfully completed races conducted in temperate and even in cool conditions.

Sickle Cell Trait Complications

Sickle cell anemia is a genetic disease of the red blood cells that causes them to become oblong in shape and inefficient at carrying oxygen. In order to functionally display the disease of sickle cell anemia, a person must have two copies of the recessive sickle cell gene. People who inherit only one copy of the gene have **sickle cell trait**. Individuals with sickle cell trait do not display the sickling of red blood cells but are at an increased risk for several pathologies that are exacerbated by exercise, dehydration, or both.

As demonstrated in a recent review of the literature, college football players who have sickle cell trait are at a 15-fold increased risk for exertional death when compared to athletes without sickle cell trait.⁴ The precise physiological mechanisms responsible for the increased risk associated with sickle cell trait are unclear; however, both defects in ryanodine receptors (see [figure 6.2](#)) and other genetic variations that affect the kidney's ability to concentrate urine and limit body water loss may contribute to this increased risk. The ryanodine receptor mediates the release of calcium from the sarcoplasmic reticulum, and mutations in this receptor are also responsible for *malignant hyperthermia*, a life-threatening condition that dramatically increases metabolic heat production during surgical anesthesia. In sickle cell trait, most sudden death events occur during high-intensity exercise plus high environmental heat and humidity, conditions that contribute to the sometimes fatal combination of severe dehydration and increased cardiovascular

strain.⁸ Sickle cell trait is now routinely screened for in National Collegiate Athletic Association student athletes,¹² an effort endorsed by many other sports medicine governing bodies.

Preventing Hyperthermia

We can do little about the prevailing weather conditions. In threatening conditions, athletes must either move the exercise session to a less stressful environment (for example, moving practice indoors) or decrease their effort (and thus their metabolic heat production) in order to reduce their risk of overheating. Athletes, coaches, and sport organizers should all be able to recognize the symptoms of heat illness.

To prevent heat disorders, several simple precautions should be taken. Competition and practice should not be held outdoors when the WBGT is more than 28 °C (82.4 °F) unless special precautions are taken. Scheduling practices and events either in the early morning or in the late evening avoids the severe heat stress of midday. Fluids should be readily available, and drink breaks should be scheduled every 15 to 30 min, with a goal of matching fluid intake to sweat loss. Because individual sweating rates and sweat sodium losses vary tremendously and cannot easily be predicted across individual athletes, athletes should customize their fluid intake based on their individual sweating rate. This is best accomplished by having athletes weigh themselves before and after exercise sessions and learn to estimate their approximate fluid needs.

Several organizations have set forth guidelines for practicing and competing under conditions of heat stress. As a summary,

1. Athletic events (distance races, tennis matches, sport team practices, and so on) should be scheduled to avoid the hottest times of the day. As a general rule, if the WBGT is above 28 °C (82-83 °F), consider cancelling, moving indoors, decreasing intensity of practice, or otherwise altering the event.
2. An adequate supply of palatable fluid must be available. Athletes should be educated and encouraged to prevent excessive (>2%) weight loss—that is, to replace their sweat

losses to prevent dehydration but not overdrink to the point where they gain weight during the event.

3. Because individual sweating rates and sweat sodium losses vary tremendously, athletes should customize their fluid intake based on their individual sweating rate. Sweating rate can be estimated by measuring body weight before and after exercise. Fluids containing electrolytes and carbohydrate can provide benefits over water alone.
4. Athletes should be aware of signs and symptoms of heat illness. Cold-water immersion is the most efficient method for cooling hyperthermic athletes in the field.
5. Organizers of events and medical personnel should have the right to cancel or terminate events and to stop individual athletes who exhibit clear signs of heat exhaustion or heatstroke.

Clothing is another important consideration. Obviously, the more clothing worn, the less body area exposed to the environment to allow for direct heat loss. The foolish practice of exercising in a rubberized suit to promote weight loss is an excellent illustration of how a dangerous microenvironment (the isolated environment inside the suit) can be created in which temperature and humidity are sufficiently high to block virtually all heat loss from the body. This can rapidly lead to heat exhaustion or heatstroke. Football uniforms are another example of clothing that impedes heat loss. Coaches and athletic trainers should avoid practice sessions in full uniforms whenever possible, especially early in the season when temperatures tend to be hottest and players tend to be less fit and not well acclimated.

Distance athletes should wear as little clothing as possible when heat stress is a potential limitation to thermoregulation. They should tend toward underdressing rather than overdressing, because the metabolic heat generated will soon make extra clothing an unnecessary burden. Clothing should be loosely woven to allow sweat to be absorbed and wicked away from the skin and should be light colored to reflect heat back to the environment. Hats should be worn during exercise in bright sunlight or when cloud cover is limited.

It is also important to maintain adequate hydration, since the body loses considerable water through sweating. This is discussed in detail in [chapter 15](#). Briefly, drinking fluid both before and during exercise can greatly reduce the negative effects of exercising in the heat. Adequate fluid intake will attenuate the increase in core body temperature and heart rate normally seen when a person exercises in the heat and will allow exercise to be continued longer. This is illustrated in [figure 12.10](#).

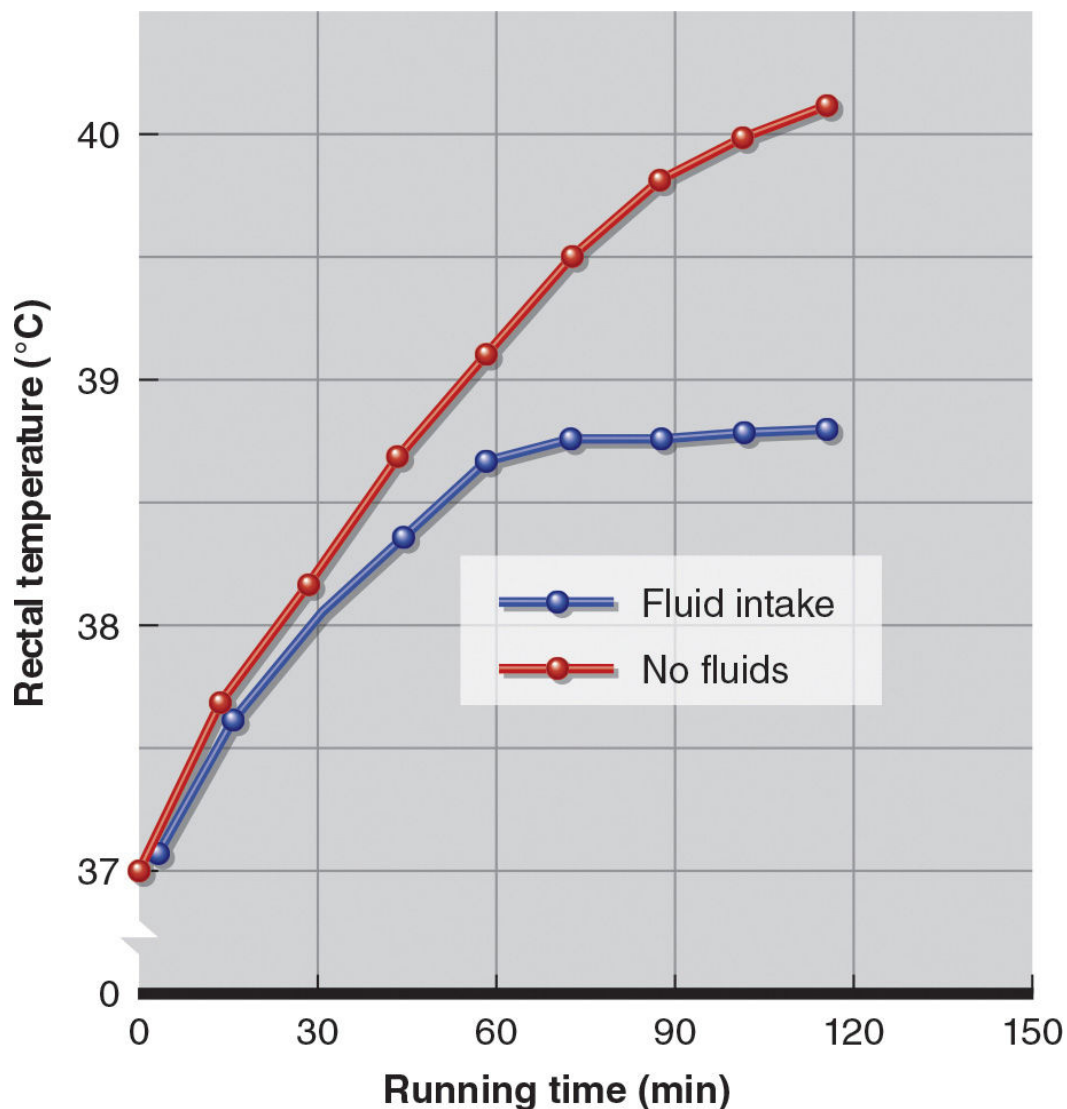


FIGURE 12.10 Effects of fluid intake on core (rectal) body temperature during a 2 h run. Subjects were given fluid during one trial, and on a separate day they completed a second trial without fluid. Note that fluid intake did not have much influence until about 45 min, after which time body heat storage was reduced compared with that in the no-fluid trial.

Based on Costill, Kammer, and Fisher (1970).

In Review

- Heat stress on the body is not accurately reflected by air temperature alone. Exercise intensity (metabolic heat), humidity, air velocity (or wind), radiation, and clothing also contribute to the total heat stress experienced during exercise in the heat.
- Perhaps the most widely used measurement of the combined physiological effects of heat stress is the WBGT, which measures air temperature and accounts for the heat exchange potential through convection, evaporation, and radiation in a specific environment. Exercise intensity and clothing must be considered separately along with the WBGT.
- To calculate WBGT:
 - Outdoor WBGT = $0.1 T_{db} + 0.7 T_{wb} + 0.2 T_g$
 - Indoor WBGT = $0.7 T_{wb} + 0.3 T_g$
- Heat cramps are caused by the loss of fluids and salt (sodium) that results from excessive sweating in susceptible athletes. High dietary sodium intakes and proper hydration strategies can prevent heat cramps.
- Heat exhaustion results from the inability of the cardiovascular system to adequately meet the blood flow needs of the active muscles and the skin. It is often brought on by dehydration caused by excessive loss of fluids and electrolytes, which results in reduced blood volume. Although it is not in itself life threatening, heat exhaustion can deteriorate to heatstroke if untreated.
- Heatstroke is caused by failure of the body's thermoregulatory mechanisms. If it is not treated, core temperature continues to rise quickly and can be fatal. In addition to a severely elevated core temperature, altered mental status or cognitive function is a hallmark sign of heatstroke.

Acclimation to Exercise in the Heat

How can athletes prepare for prolonged activity in the heat? Does repeated exercise in the heat make us better able to tolerate thermal stress? Many studies have addressed these questions and have concluded that repeated exercise in the heat causes a series of relatively rapid adaptations that enable us to perform better, and more safely, in hot conditions. When these physiological changes occur over short periods of time, like days to weeks, or if they are artificially induced as in a climatic chamber, those adaptations are

termed **acclimation (heat acclimation)**. A similar but much more gradual set of adaptations occurs in people who adapt to hot conditions by living in hot environments for months to years. This is known as **acclimatization** (note that the word *climate* is part of this latter term).

Effects of Heat Acclimation

Repeated bouts of prolonged, low-intensity exercise in the heat cause a relatively rapid improvement in the ability to maintain cardiovascular function and eliminate excess body heat, which reduces physiological strain. This process, termed heat acclimation, involves changes in plasma volume, cardiovascular function, sweating, and skin blood flow that allow for subsequent exercise bouts in the heat to be performed with a lower core temperature and heart rate response (figure 12.11). Because the body's heat loss capacity at a given rate of work is enhanced by acclimation, core temperature during exercise increases less than before acclimation (figure 12.11a), and heart rate increases less in response to standardized submaximal exercise after heat acclimation (figure 12.11b). In addition, after heat acclimation, more work can be done before adverse symptoms occur or a maximal tolerable core temperature or heart rate is reached.

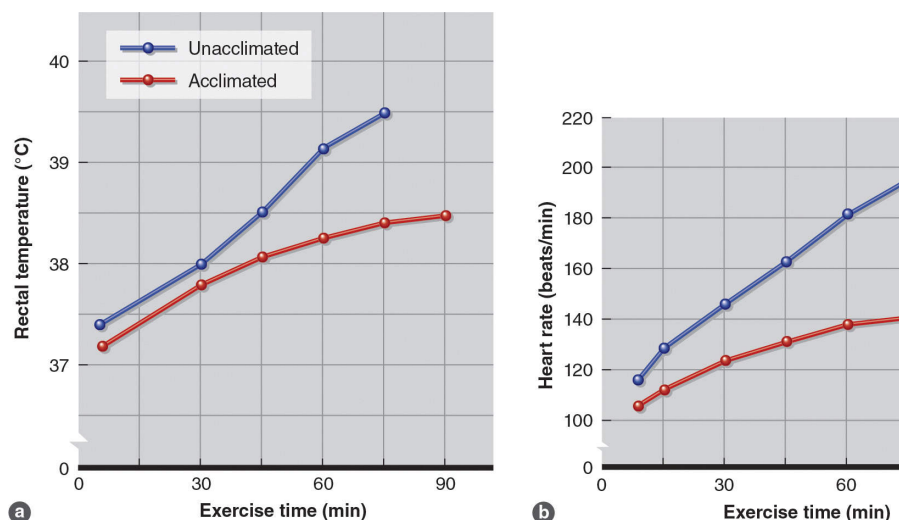


FIGURE 12.11 Typical (a) rectal temperature and (b) heart rate responses during an acute bout of exercise at the same intensity before and after heat acclimation. Note that, in addition to lower physiological strain, exercise time is usually longer after acclimation.

Data from King et al. (1985).

This series of positive adaptations typically takes a period of 9 to 14 days of exercise in the heat to fully occur, as shown in [figure 12.12](#). Well-trained individuals need fewer exposures than untrained individuals to fully acclimate. A critical physiological adjustment that occurs over the first 1 to 3 days of acclimation is the expansion of plasma volume. The exact mechanism by which plasma volume expands after these initial exercise–heat exposures is not universally agreed upon. The process likely involves (1) proteins being forced out of the circulation as muscles contract, (2) these same proteins then being returned to the blood through the lymph, and (3) fluid moving into the blood because of the oncotic pressure exerted by the increased protein content. However, this change is temporary, and blood volume usually returns to original levels within 10 days. This early expansion of blood volume is important because it supports stroke volume, allowing the body to maintain cardiac output while additional physiological adjustments are made.

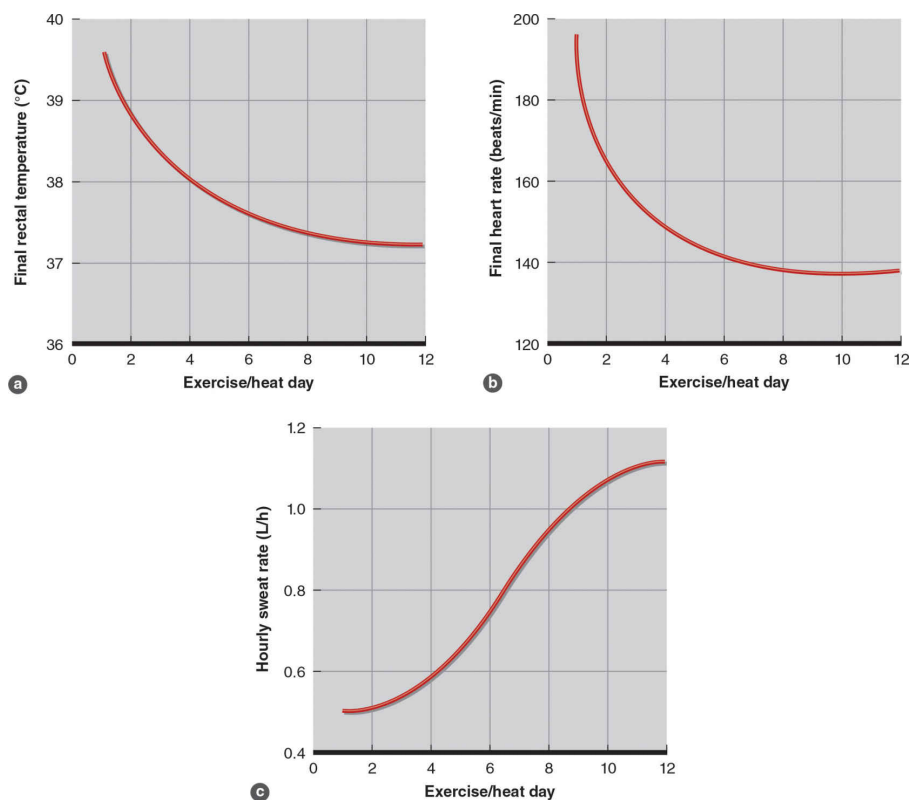


FIGURE 12.12 Changes in (a) rectal temperature, (b) heart rate, and (c) hourly sweating rate in a group of men exercising for 100 min per day for 12 consecutive days in a hot environment.

As shown in [figure 12.12](#), end-exercise heart rate and core temperature decrease early in the acclimation process, while the increase in sweating rate during exercise in the heat occurs somewhat later. An additional adaptation is a more even distribution of sweat over the body, with increased sweating on the most exposed body areas such as the arms and legs, areas that are most effective at dissipating body heat. At the beginning of exercise, sweating also starts earlier in an acclimated person, which improves heat tolerance, and the sweat that is produced becomes more dilute, conserving sodium. This latter effect occurs in part because the eccrine sweat glands become more sensitive to the effects of circulating aldosterone.

Achieving Heat Acclimation

Heat acclimation requires more than merely resting in a hot environment. The benefits of acclimation, as well as the rate at which we acclimate, depend on

- the environmental conditions during each exercise session,
- the duration of exercise–heat exposure, and
- the rate of internal heat production (exercise intensity).

An athlete must exercise in a hot environment to attain full acclimation that sustains exercise in the heat. Simply sitting in a hot environment, such as a sauna or steam room, for long periods each day will not fully or adequately prepare the individual for physical exertion in the heat, at least not to the same extent as will exercising in the heat.

Because body temperature is elevated and sweating occurs, athletes gain partial heat tolerance simply by training, even in a cooler environment. Therefore, athletes are “preacclimated” to heat and need fewer exercise heat exposures to fully acclimate. To gain maximal benefits, athletes who train in environments cooler than those in which they will compete must achieve heat acclimation before the contest or event. Heat acclimation will improve their performance and reduce the associated physiological stress and risk of heat injury.

With a full 14-day heat acclimation schedule (90 min cycling, 40 °C, 20% relative humidity), improvements in whole-body heat loss can be demonstrated using direct calorimetry.⁹ Direct calorimetry is described in [chapter 5](#). After 14 days, whole-body heat loss was improved by 11% since sweat rate, and subsequently sweat evaporation, occurred to a greater extent over a shorter period of time. Approximately 70% of the improvements occurred within the first 7 days. While heat loss responses improved, there was no effect on core temperature in this study. Once the acclimation protocol was stopped, benefits persisted for up to 2 weeks.

Sex Differences

Exercise in the heat, in the cold, or at altitude confers additional stress or challenge to the body's adaptive abilities. Many early studies indicated that women are less tolerant to heat than men are, particularly when exercising. Much of this difference, however, was the result of lower fitness levels of the women included in these studies, because the men and women were tested at the same absolute rate of work. When the rate of work is adjusted relative to individual $\dot{V}O_{2\max}$ values, women's responses are almost identical to men's. Women have a delayed onset of sweating and dilation of the skin arterioles during the luteal phase of the menstrual cycle (i.e., onset occurs at a higher core temperature). However, this should not affect performance until core temperature approaches 40 °C. Women generally have lower sweat rates for the same exercise and heat stress. Although they possess a larger number of active sweat glands than men do, women produce less sweat per gland. This is a slight disadvantage in hot, dry environments but is a slight advantage in humid conditions in which sweat evaporation is minimal and reduced sweating delays dehydration.

After acclimatization, the internal temperature at which sweating and vasodilation begin is lowered to a similar extent in women and men. Also, the sensitivity of the sweating response per unit increase in internal temperature increases by a similar amount in the two sexes following both physical training and heat acclimatization. Therefore, most differences noted between women and men in the

early studies can be attributed to initial differences in their physical conditioning and acclimation status and not to their sex.

In Review

- Repeated exercise exposure to heat stress gradually improves one's ability to support cardiovascular function and lose excess heat during subsequent bouts of exercise heat stress. This process is called heat acclimation.
- With heat acclimation, people start to sweat earlier and sweating rate increases, particularly on areas that are well exposed and are the most efficient at promoting heat loss. This reduces skin temperature, which increases the thermal gradient from the skin to the environment and promotes heat loss.
- Both core temperature and heart rate during exercise in the heat are reduced at any given intensity after heat acclimation. Plasma volume increases early in the process, contributing to an increase in stroke volume that supports the delivery of blood to both active muscles and skin.
- Full heat acclimation requires exercise in a hot environment, not merely exposure to heat.
- The rate of heat acclimation depends on training status, the conditions to which one is exposed during each session, the duration of the exposure, and the rate of internal heat production.
- With moderate-intensity exercise in the heat, people generally acclimate in 9 to 14 exposure days. Cardiovascular adjustments, which depend on expansion of plasma volume, occur first, followed by changes in sweating.
- When exercise intensity is adjusted relative to an individual's $\dot{V}O_{2\max}$, women and men respond similarly to heat stress. Most differences noted are likely attributable to different initial levels of conditioning. Women tend to have lower sweating rates, which is advantageous in hot, humid conditions and disadvantageous in hot, dry conditions.

Exercise in the Cold

Humans can be thought of as tropical animals. Most of our adjustments to heat stress are physiological, whereas many adjustments to cold environments involve behavior, like putting on more clothing or seeking shelter. Increased year-round participation in sport has sparked new interest in, and concerns about, exercise in the cold. In addition, certain occupations and military endeavors

require people to work in cold conditions—conditions that often limit performance. For these reasons, the physiological responses and health risks associated with cold stress are important issues in exercise science. We define cold stress here as any environmental condition causing a loss of body heat that threatens homeostasis. In the following discussion we focus on the two major cold environments: air and water.

The hypothalamus has a temperature set point of about 37 °C (98.6 °F), but daily fluctuations in the body temperature can be as much as 1 °C. A decrease in either skin or blood temperature provides feedback to the thermoregulatory center (POAH) to activate mechanisms that conserve body heat and increase heat production. The primary means by which our bodies avoid excessive heat loss (in the order in which they are invoked) are peripheral vasoconstriction, nonshivering thermogenesis, and shivering. Because these mechanisms or effectors of heat production and conservation are often inadequate, we also must rely on behavioral responses such as huddling behavior (decreasing exposed body surface area) and putting on more clothing to help insulate our deep body tissues from the environment.

Peripheral vasoconstriction occurs as a result of sympathetic stimulation of the smooth muscle layers of the arterioles in the skin. This stimulation causes the vascular smooth muscle to contract, which constricts the arterioles, reduces the blood flow to the shell of the body, and minimizes heat loss. Even at thermoneutral temperatures, there is tonic (constant baseline) skin vasoconstriction, and continuous adjustment of skin vascular tone occurs at all times to offset small heat imbalances in the body. When changing skin blood flow alone is not adequate to prevent heat loss, **nonshivering thermogenesis**—stimulation of metabolism by the SNS—is increased. Increasing the metabolic rate increases heat production. The next line of defense of body temperature during cold stress is **shivering**, a rapid, involuntary cycle of contraction and relaxation of skeletal muscles, which can cause a four- to fivefold increase in the body's rate of heat production. The overall adjustments in blood flow and metabolism that serve to maintain body core temperature are shown in [figure 12.13](#).

Habituation and Acclimation to Cold

Whether humans truly acclimate to cold environments in the physiological sense—and if so, how this occurs—is far less clear than the process of heat acclimation. When one looks at studies of people who undergo repeated daily cold exposures, the results seem controversial. However, Dr. Andrew Young of the U.S. Army Research Institute for Environmental Medicine and others have proposed a scheme to explain the development of different patterns of cold adaptation that are observed in humans.¹⁶ People who are regularly exposed to cold environments under which significant body heat loss does not occur typically undergo **cold habituation**, in which skin vasoconstrictor and shivering responses are blunted, and core temperature is allowed to fall to a greater extent than before the chronic cold exposures. This pattern of adaptation often occurs when small areas of skin—often the hands and face—are exposed repeatedly to cold air.

However, when heat loss is more severe or occurs at a faster rate, total body heat loss may occur. In cases in which increased metabolic heat production alone can sufficiently minimize heat loss, enhanced nonshivering and shivering thermogenesis develop (**metabolic acclimation**). Yet a third distinct pattern of cold adaptation, called **insulative acclimation**, tends to occur in situations in which increased metabolism is unable to maintain core temperature. In insulative acclimation, enhanced skin vasoconstriction occurs, which increases peripheral insulation and minimizes heat loss.

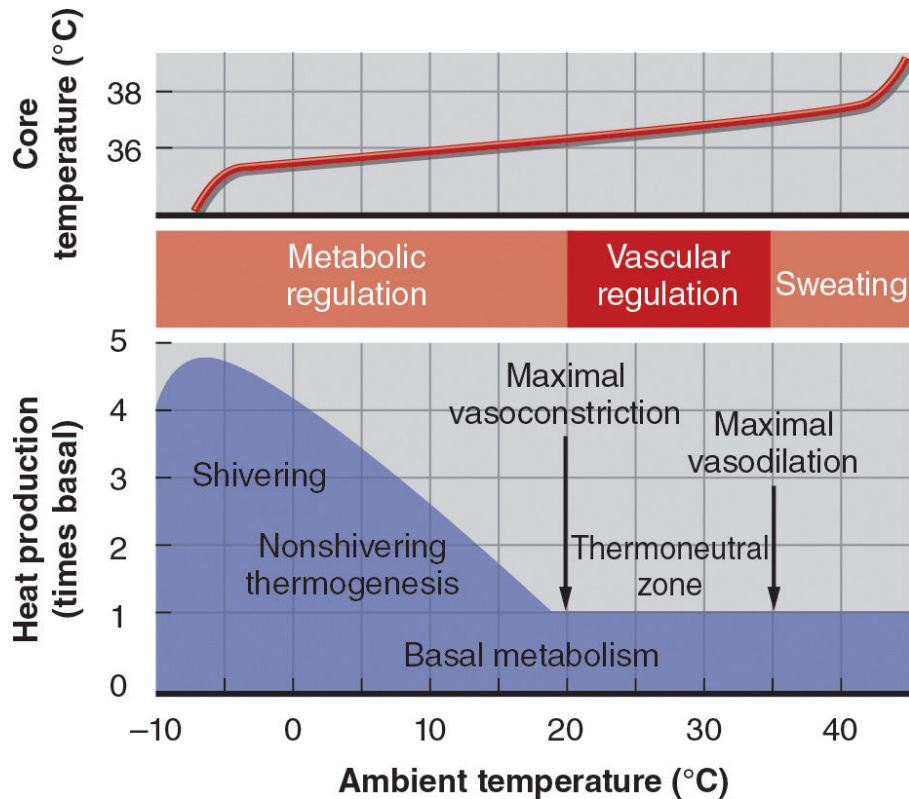


FIGURE 12.13 Thermoregulatory mechanisms by which humans strive to maintain a relatively constant body core temperature. In the thermoneutral zone, minor adjustments to skin blood flow minimize heat loss or gain. When maximal vasoconstriction is not sufficient to maintain core temperature, metabolic regulation, first in the form of nonshivering thermogenesis and then through shivering, serves to increase metabolic heat production.

Other Factors Affecting Body Heat Loss

The mechanisms of conduction, convection, radiation, and evaporation, which usually perform effectively in dissipating metabolically produced heat during exercise in warm conditions, can dissipate heat to the environment faster than the body produces it in extremely cold environments.

RESEARCH PERSPECTIVE 12.3

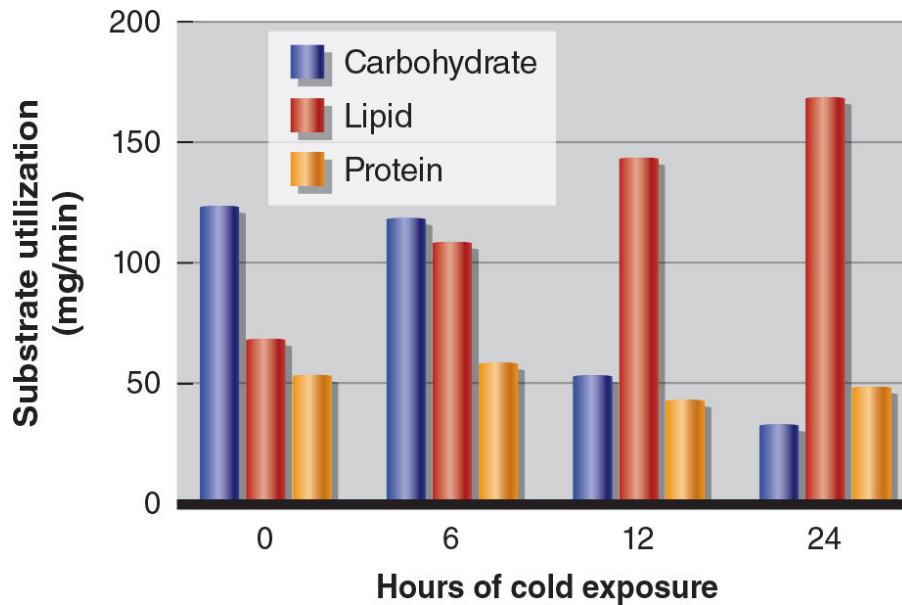
Fuel for Shivering

During environmental cold exposure when behavioral changes (e.g., putting on additional layers, moving to a warmer area) are impossible or ill advised, humans rely on involuntary processes such as shivering to increase metabolic heat production. The muscle contractions associated with shivering are fueled by the oxidation of carbohydrates, lipids, and proteins, with the precise mix of fuels determined by the availability of each substrate and the

muscle recruitment patterns and intensity of the shivering. Adult humans are able to sustain shivering and the metabolic heat production associated with it using a wide variety of fuels over a span of a few hours, and research has shown that when one fuel store is reduced, utilization of other fuels is increased to maintain shivering thermogenesis. Until recently, detailed studies of fuel sources for shivering had never been conducted in trials lasting longer than 3 h. Because of this, the long-term metabolic demands of cold exposure were largely unknown, and it was unclear if substrate depletion could limit shivering thermogenesis during extended cold exposures, such as during survival situations.

Recently, a team of research scientists from Canada conducted a 24 h cold-survival simulation to study the oxidative fuel selection of shivering during prolonged cold exposure in underfed and noncold-acclimatized men.³ In this study, participants came to the laboratory in the morning following an overnight fast and were dressed in light cotton clothing, fleece mittens, and neoprene boots with their extremities protected to minimize the risk of nonfreezing cold injury. The subjects swallowed a telemetry pill to measure core temperature throughout the experiment and were fitted with thermal and electromyography sensors to measure skin temperature and muscle contractions. After baseline measurements, participants completed a 5 km treadmill walk to simulate physical activities associated with survival in an emergency. The subjects were then moved to an environmental chamber where they were exposed to 7.5 °C (45.5 °F) and 50% relative humidity for 24 h. Over the 24 h cold exposure, the subjects were fed 1,641 kcals divided into rations provided at 0, 3, 6, 9, 12, 15, 18, and 21 h. Metabolic and thermal responses as well as electromyography activity were measured in a seated position for 1 h at 6, 12, and 24 h of cold exposure. Changes in thermogenesis, fuel selection, and fuel availability were measured periodically using indirect calorimetry and metabolic isotope tracers.

Cold exposure decreased core temperature ~ 0.8 °C (1.4 °F) and mean skin temperature ~6.1 °C (11 °F). Total heat production initially increased 1.3 to 1.5 times compared to that at baseline and remained constant throughout the trial. This rise in metabolic heat production and shivering intensity was fueled by large modifications in fuel selection throughout the cold exposure. Total carbohydrate oxidation decreased after 6 h and remained low while lipid oxidation progressively doubled by 24 h (see figure). Protein utilization was low and did not change throughout the protocol. The research team concluded that these changes in fuel selection over an extended period of cold exposure dramatically reduced the utilization of limited muscle glycogen stores, extending the predicted time to muscle glycogen depletion as much as 10 times. Future research will be necessary to determine how nutritional substrate delivery or more extreme cold exposure may affect these responses.



Representation of fuel sources for shivering thermogenesis after 0, 6, 12, and 24 h of cold exposure (7.5 °C, 50% relative humidity) in an environmental chamber. Carbohydrate oxidation decreased after 6 h while lipid oxidation progressively increased over the 24 h exposure. Protein utilization remained mostly unchanged throughout.

Pinpointing the exact conditions that permit excessive body heat loss and eventual **hypothermia** (low body core temperature) is difficult. Thermal balance depends on a wide variety of factors that affect the balance between body heat production and heat loss. Generally speaking, the larger the thermal gradient between the skin and the cold environment, the greater the heat loss. When exercising in the cold, people should not overdress. Overdressing can cause the body to become hot and initiate sweating. As the sweat soaks through the clothing, evaporation removes the heat, and heat is lost at an even faster rate.

A number of additional factors can also influence the rate of heat loss.

Body Size and Composition

Insulating the body against the cold is the most obvious form of protection against hypothermia. Recall that insulation is defined as resistance to dry heat exchange through radiation, convection, and conduction. Both inactive peripheral muscles and subcutaneous fat are excellent insulators. Skinfold measurements of subcutaneous fat

thickness are a good indicator of an individual's tolerance for cold exposure. The thermal conductivity of fat (its capacity for transferring heat) is relatively low, so fat impedes heat transfer from the deep tissues to the body surface. People who have more fat mass conserve heat more efficiently in the cold than smaller, leaner individuals.

The rate of heat loss also is affected by the ratio of body surface area to body mass. Larger individuals have a small surface area-to-mass ratio, which makes them less susceptible to hypothermia.

Sex and Age Differences

Women have a slight advantage over men during cold exposure because they have more subcutaneous body fat. But their smaller muscle mass is a disadvantage in extreme cold because shivering is the major adaptation for generating body heat. The greater the active muscle mass, the greater the heat generation. Muscle also provides an additional insulating layer.

True sex differences in cold tolerance are minimal. Some studies have shown that the added subcutaneous fat in women may give them an advantage during cold-water immersion, but when men and women of similar body fat mass and size are compared, no real difference is noted in body temperature regulation during exposure to the cold.

As shown in [table 12.2](#), small children have a large surface area-to-mass ratio compared with adults, leading to proportionately greater heat loss. This makes it more difficult for them to maintain normal body temperature in the cold. On the other end of the age spectrum, elderly individuals often tend to lose overall muscle mass, decreasing tissue insulation and making them more susceptible to hypothermia.

Windchill

As with heat, the air temperature alone does not provide a valid index of the heat loss experienced by the individual. Air movement, or wind, increases convective heat loss and therefore increases the rate of cooling. **Windchill** is an index based on the cooling effect of wind and is an often misunderstood and misused concept. Windchill is typically presented in charts of windchill equivalent temperatures

showing various combinations of air temperature and wind speed that result in the same cooling power as that seen with no wind (figure 12.14). It is important to remember that windchill is not the temperature of the wind or the air (windchill does *not* change air temperature). True windchill refers to the cooling power of the environment. As windchill increases, so does the risk of freezing of tissues (figure 12.14).

Heat Loss in Cold Water

More research has been conducted on cold-water exposure than on exposure to cold air. Whereas radiation and sweat evaporation are the primary mechanisms for heat loss in air, convection allows the greatest heat transfer during immersion in water (recall that convection involves heat loss to moving fluids, which include both liquids and gases). As mentioned earlier, water has a thermal conductivity about 26 times greater than air. This means that heat loss by convection is 26 times faster in cold water than in cold air. When all heat-transfer mechanisms are considered (radiation, conduction, convection, and evaporation), the body generally loses heat four times faster in water than it does in air of the same temperature.

TABLE 12.2 Body Weight, Height, Surface Area, and Surface Area/Mass Ratio for an Average-Sized Adult and Child

Person	Weight (kg)	Height (m)	Surface area (m ²)	Area/mass ratio (m ² /kg)
Adult	85	1.83	2.07	0.024
Child	25	1.00	0.79	0.032

At rest, humans generally maintain a constant internal temperature in water at temperatures down to about 32 °C (89.6 °F). But when the water temperature decreases further, they may become hypothermic. Because of the large loss of heat from a body immersed in cold water, prolonged exposure or unusually cold water can lead to extreme hypothermia and death. Individuals immersed in water at 15 °C (59 °F) experience a decrease in rectal temperature of about 2.1 °C (3.8 °F) per hour. In 1995, four U.S. Army rangers died of hypothermia after exposure to 11 °C (52 °F) swamp water in

Florida, tragically publicizing the fact that hypothermia can occur when water and air temperatures are well above freezing.

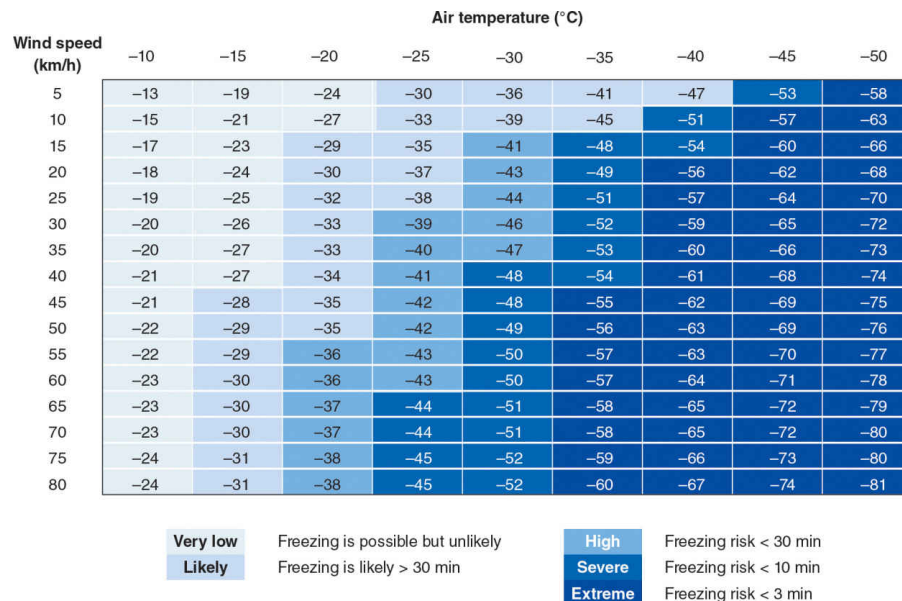


FIGURE 12.14 Windchill equivalent temperature chart showing various combinations of temperature and wind speed that result in the same cooling power as that seen with no wind. For example, a wind speed of 20 km/h at -10°C would result in the same heat loss as -30°C with no wind. Also shown in the figure is the risk of tissues freezing as windchill—the cooling power of the environment—increases.

If the water temperature were lowered to 4°C (39.2°F), rectal temperature would decrease at a rate of 3.2°C (5.8°F) per hour. The rate of heat loss is further accelerated if the cold water is moving around the individual because heat loss by convection increases. As a result, survival time in cold water under these conditions is quite brief. Victims can become weak and lose consciousness within minutes.

If the metabolic rate is low, as when the person is at rest, then even moderately cool water can cause hypothermia. But exercise in water increases the metabolic rate and offsets some of the heat loss. For example, although heat loss increases when one is swimming at high speeds (because of convection), the swimmer's accelerated rate of metabolic heat production more than compensates for the greater heat transfer. For competition and training, water temperatures between 23.9 and 27.8°C (75 - 82°F) are appropriate.

Physiological Responses to Exercise in the Cold

We have seen how the body adapts to maintain its internal temperature when exposed to a cold environment. Now consider what happens when the demands of physical performance are added to those of thermoregulation in the cold. How does the body respond to exercise in cold environmental conditions?

Muscle Function

Cooling a muscle causes it to contract with less force. The nervous system responds to muscle cooling by altering the normal muscle fiber recruitment patterns for force development, which may decrease the efficiency of the muscle's actions. Both muscle shortening velocity and power decrease significantly when muscle temperature is lowered. Luckily, large deep muscles seldom experience such low temperatures because they are protected from heat loss by a continuous supply of warm blood flow.

If clothing insulation and exercise metabolism are sufficient to maintain an athlete's body temperature in the cold, aerobic exercise performance may be unimpaired. However, as fatigue sets in and intensity decreases, so does metabolic heat production. Long-distance running, swimming, and skiing in the cold can expose the participant to such conditions. At the beginning of these activities, the athlete can exercise at an intensity that generates sufficient metabolic heat to maintain core temperature. However, late in the activity, when the energy reserves have diminished, exercise intensity declines, and this reduces metabolic heat production. The resulting decrease in core temperature causes the individual to become even more fatigued and less capable of generating heat. Under these conditions, the athlete may be confronted with a potentially dangerous situation.

Cold conditions affect muscle function in another way. As small muscles in the periphery like the fingers become cold, muscle function can be severely affected. This results in a loss of manual dexterity and limits the ability to perform fine motor skills like writing and manual labor tasks.

Metabolic Responses

Prolonged exercise increases the mobilization and oxidation of free fatty acids (FFAs) as a fuel source. The primary stimulus for this increased lipid metabolism is the release of catecholamines (epinephrine and norepinephrine). Exposure to cold markedly increases epinephrine and norepinephrine secretion, but FFA levels increase substantially less than during prolonged exercise in warmer conditions. Cold exposure triggers vasoconstriction in the vessels supplying not only the skin but also fatty subcutaneous tissues as well. The subcutaneous fat is a major storage site for lipids (as adipose tissue), so this vasoconstriction reduces the blood flow to an important area from which the FFAs would be mobilized. Thus, FFA levels do not increase as much as the elevated levels of epinephrine and norepinephrine would predict.

Blood glucose plays an important role in both cold tolerance and exercise endurance. For example, hypoglycemia (low blood sugar) suppresses shivering. The reasons for these changes are unknown. Fortunately, blood glucose concentrations are maintained reasonably well during cold exposure. Muscle glycogen, on the other hand, is used at a somewhat higher rate in the cold than in warmer conditions. However, studies on exercise metabolism in the cold are limited, and our knowledge regarding hormonal regulation of metabolism in the cold is too limited to support any definitive conclusions.



In Review

- Peripheral vasoconstriction decreases the transfer of heat from the skin to the air, thus decreasing heat loss to the environment. This is the body's first line of defense in the cold.
- The body's insulating shell consists of two regions: the superficial skin and subcutaneous fat and the underlying muscle. Increased skin vasoconstriction, increased subcutaneous fat thickness, and increased inactive muscle mass, especially in the limbs, can all increase total body insulation.
- Nonshivering thermogenesis increases metabolic heat production through activation of the SNS and by the action of hormones. Shivering thermogenesis

increases metabolic heat production further to help maintain or increase body temperature.

- There are three distinct patterns of adaptation to repeated cold exposure: cold habituation, metabolic acclimation, and insulative acclimation.
 - Body size is an important consideration for heat loss. Both a higher ratio of surface area to mass and a lower peripheral muscle mass or subcutaneous fat increase the loss of body heat to the environment.
 - Because they have more insulating subcutaneous fat, women have a slight advantage over men during cold exposure, but their smaller muscle mass limits their ability to generate body heat.
 - Wind increases heat loss by convection. The cooling power of the environment, known as windchill, is typically expressed as equivalent temperatures.
 - Immersion in cold water tremendously increases heat loss through convection. In some cases, exercise may generate enough metabolic heat to offset some of this loss.
 - When muscle is cooled, it is less able to produce force, and fatigue can occur more rapidly.
 - During prolonged exercise in the cold, as fatigue causes exercise intensity to decline, metabolic heat production decreases and exercisers may become susceptible to hypothermia.
 - Exercise triggers release of catecholamines, which increase the mobilization and use of FFAs for fuel. But in the cold, vasoconstriction impairs circulation to peripheral fat stores, so this process is attenuated.
-

Health Risks During Exercise in the Cold

If humans had retained the ability of lower animals, such as reptiles, to tolerate low body temperatures, we could survive extreme hypothermia. Unfortunately, the evolution of thermoregulation in humans has been accompanied by a diminished ability of tissues to function effectively outside a narrow range of temperatures. This section deals briefly with the health risks associated with cold stress. The American College of Sports Medicine published a comprehensive position stand, "Prevention of Cold Injuries During Exercise," in 2006 that addresses these topics in much greater detail.¹



Hypothermia

Individuals immersed in near-freezing water will die within a few minutes when their rectal temperature decreases from a normal level of 37 °C (98.6 °F) to 24 or 25 °C (75.2 or 77 °F). Cases of accidental hypothermia, as well as data obtained from surgical patients who are intentionally made hypothermic, reveal that the lethal lower limit of body temperature is usually between 23 and 25 °C (73.4-77 °F), although patients have recovered after having rectal temperatures below 18 °C (64.4 °F).

RESEARCH PERSPECTIVE 12.4

The Yukon Arctic Ultramarathon

The Yukon Arctic Ultramarathon is considered the longest and coldest ultra-endurance event in the world. This 430 mi (692 km) event, which follows the Yukon Quest Trail beginning in Whitehorse (the capitol of the Canadian province of Yukon), is made up a series of nonstop multiday races of mountain biking, cross-country skiing, or walking. Moreover, it is run in February of each year. All participants must be self-reliant and pull a sled or attach a pack to their bike that contains all provisions and gear (including food, water, and tent) needed for the event. Because of the grueling conditions caused by the time of year and the extreme length, the Yukon Arctic Ultra is often ranked as the most difficult physical competition in the world. Cold exposure and exercise have been reported to influence circulating concentrations of myokines, adipokines, and hepatokines that may

alter the regulation of metabolism. Specifically, changes in these metabolic signaling molecules may initiate changes in white or brown adipose tissue to promote thermogenesis.

In 2016, a team of research scientists from the University of Alaska Fairbanks and the Center for Space Medicine and Extreme Environments in Berlin, Germany, followed eight participants in the Yukon Arctic Ultramarathon.² Blood samples were collected at preevent, midevent, and postevent checkpoints. The temperature during the event ranged from -45°C (-49°F) to -8°C (17.6°F) that year. Because of these exceptionally harsh conditions, four of the eight study participants withdrew from competition by the 300 mi (483 km) mark. The four who did pass the 300 mi mark all lost significant body weight and fat mass. Analysis of the blood samples collected indicated that serum **irisin**, a myokine that has been purported to promote the breakdown of lipids and support brown adipose tissue formation, increased over the event. However, there were no changes in other myokines, adipokines, or hepatokines, and serum metabolites remained stable throughout the event. The data from the four participants who finished the Yukon Arctic Ultramarathon provide a glimpse into the physiology of these athletes and may suggest that the combination of cold exposure and extreme physical exertion promotes alterations in substrate metabolism that preserve skeletal muscle and increase physiological resilience in the face of these excessive demands. In the future, these researchers plan to obtain measurements of energy expenditure, dietary intake, and sleep quality to determine their contribution to the relative preservation of lean body mass in these athletes.

Once core temperature falls below about 34.5°C (94.1°F), the hypothalamus begins to lose its ability to regulate body temperature. This ability is completely lost when the internal temperature decreases to about 29.5°C (85.1°F). This loss of function is associated with a slowing of metabolic reactions. For each 10°C (18°F) drop in cellular temperature, the metabolism of the cell decreases by half. As a result, low core temperatures can cause drowsiness, lethargy, and even coma.

Mild hypothermia can be treated by removing the affected person from the cold, providing dry clothing and blankets for insulation, and providing warm beverages. Moderate to severe cases of hypothermia require gentle handling to avoid initiating a cardiac arrhythmia. This requires slowly rewarming the victim. Severe cases of hypothermia require hospital facilities and medical care. Recommendations to prevent cold exposure–related injuries were

outlined by the American College of Sports Medicine in its 2006 position stand on cold illness during exercise.¹

Cardiorespiratory Effects

The hazards of excessive cold exposure include potential injury to both peripheral tissues and the cardiovascular and respiratory systems. Death from hypothermia has resulted from cardiac arrest while respiration was still functional. Cooling primarily influences the sinoatrial node, the heart's pacemaker, leading to a substantial decrease in heart rate and, ultimately, cardiac arrest.

People have questioned whether rapid, deep breathing of cold air can damage or freeze the respiratory tract. In fact, the cold air that passes into the mouth and trachea is rapidly warmed, even when the temperature of inhaled air is less than -25°C (-13°F).⁶ Even at this temperature, when a person is at rest and breathing primarily through the nose, the air is warmed to about 15°C (59°F) by the time it has traveled about 5 cm (2 in.) into the nasal passage. As shown in [figure 12.15](#), extremely cold air entering the nose is quite warm by the time it reaches the back of the nasal passage, thereby posing no threat of damage to the throat, trachea, or lungs. Mouth breathing, which often occurs during exercise, may result in cold irritation to the mouth, pharynx, trachea, and even bronchi when the air temperature is below -12°C (10°F). Excessive cold exposure also affects respiratory function by decreasing respiratory rate and volume.

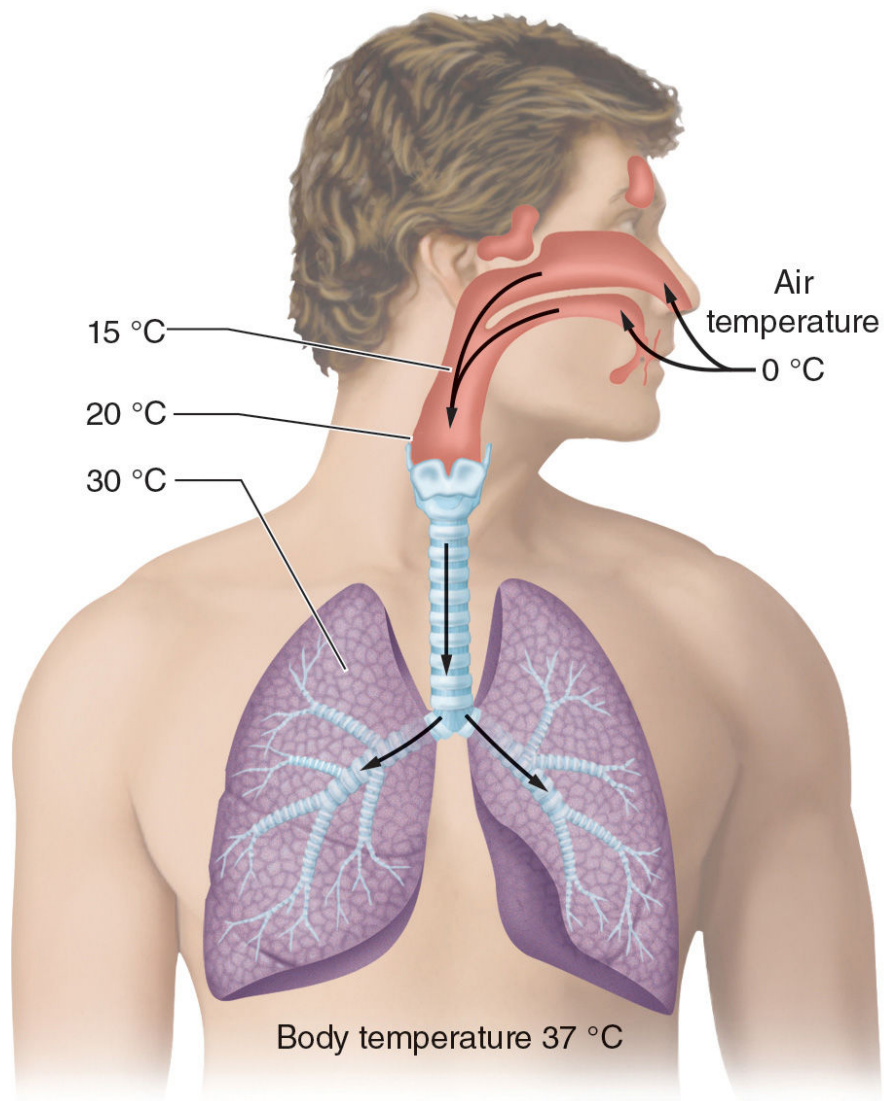


FIGURE 12.15 The warming of inspired air as it moves through the respiratory tract.



Frostbite

Exposed skin can freeze when its temperature is lowered just a few degrees below the freezing point (0 °C, or 32 °F). Because of the warming influence of circulation and metabolic heat production, the environmental air temperature (including windchill; see [figure 12.14](#)) required to freeze exposed fingers, nose, and ears is about -29 °C (-20 °F). Recall from our earlier discussion that peripheral vasoconstriction helps the body retain heat. Unfortunately, during exposure to extreme cold, the circulation in the skin can decrease to the point that the tissue dies from lack of oxygen and nutrients. This is commonly called **frostbite**. If not treated early, frostbite injuries can be serious, leading to gangrene and loss of tissue. Frostbitten parts should be left untreated until they can be thawed, preferably in a hospital, without risk of refreezing.

Exercise-Induced Asthma

While not strictly a cold-related illness, exercise-induced asthma is a common problem that affects as many as 50% of winter sport athletes (see [chapter 8](#)). The main cause of this syndrome is the drying of the airways due to the combination of the high respiration rate associated with exercise and the extremely dry air as temperature drops, although multiple environmental and genetic factors may determine whether a given athlete will experience asthma symptoms as a result. The resultant airway narrowing often leaves athletes gasping for breath. Luckily, there are preventive medications available such as β -agonists plus inhalers that can quickly deliver corticosteroids and bronchodilating medications to relieve symptoms.

In Review

- The hypothalamus begins to lose its ability to regulate body temperature when core temperature drops below about 34.5 °C (94.1 °F).
- Hypothermia critically affects the heart's sinoatrial node, decreasing the heart rate, which in turn reduces cardiac output.
- Breathing cold air does not freeze the respiratory passages or the lungs because the inspired air is progressively warmed as it moves through the respiratory tract.
- Exposure to extreme cold decreases respiratory rate and volume.

- Frostbite occurs as a consequence of the body's attempt to prevent heat loss by skin vasoconstriction. If vasoconstriction is prolonged, the skin cools rapidly and the reduced blood flow, combined with the lack of oxygen and nutrients, ultimately can cause cutaneous tissue to die.
 - Because cold air is inherently dry, many athletes experience the symptoms of exercise-induced asthma during high-intensity exercise in cold environments.
-

IN CLOSING

In this chapter, we began our examination of how the external environment affects the body's ability to perform physical work. We looked at the effects of extreme heat and cold stress and the body's responses to these conditions. We considered the health risks associated with these temperature extremes and how the body adapts to these conditions through acclimation. In the next chapter, we examine additional environmental extremes associated with exercise at altitude.

KEY TERMS

acclimation (heat acclimation)
acclimatization
arginine vasopressin
cold habituation
conduction (K)
convection (C)
critical temperature theory
dry heat exchange
eccrine sweat glands
evaporation (E)
frostbite
heat cramps
heat exhaustion
heatstroke
hypothermia
insulation
insulative acclimation
irisin
metabolic acclimation
nonshivering thermogenesis
peripheral vasoconstriction
preoptic-anterior hypothalamus (POAH)
radiation (R)
shivering
sickle cell trait
thermal stress
thermoreceptors
thermoregulation

vasopressin

wet-bulb globe temperature (WBGT)

windchill

STUDY QUESTIONS

1. What are the four major avenues for loss of body heat?
2. Which of these four pathways is most important for controlling body temperature at rest? During exercise?
3. What happens to the body temperature during exercise and why?
4. Why is water vapor pressure in the air an important factor when one is performing in the heat? Why are wind and cloud cover important?
5. What factors may limit the ability to continue to exercise in hot environments?
6. What is the purpose of the wet-bulb globe temperature (WBGT)? What does it measure?
7. Differentiate between heat cramps, heat exhaustion, and heatstroke.
8. What physiological adaptations occur that allow a person to acclimate to exercise in the heat?
9. How does the body minimize excessive heat loss during cold exposure?
10. What are the three patterns of cold adaptation and when might each be expected to occur?
11. What factors should be considered to provide maximal protection when people are exercising in the cold?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

